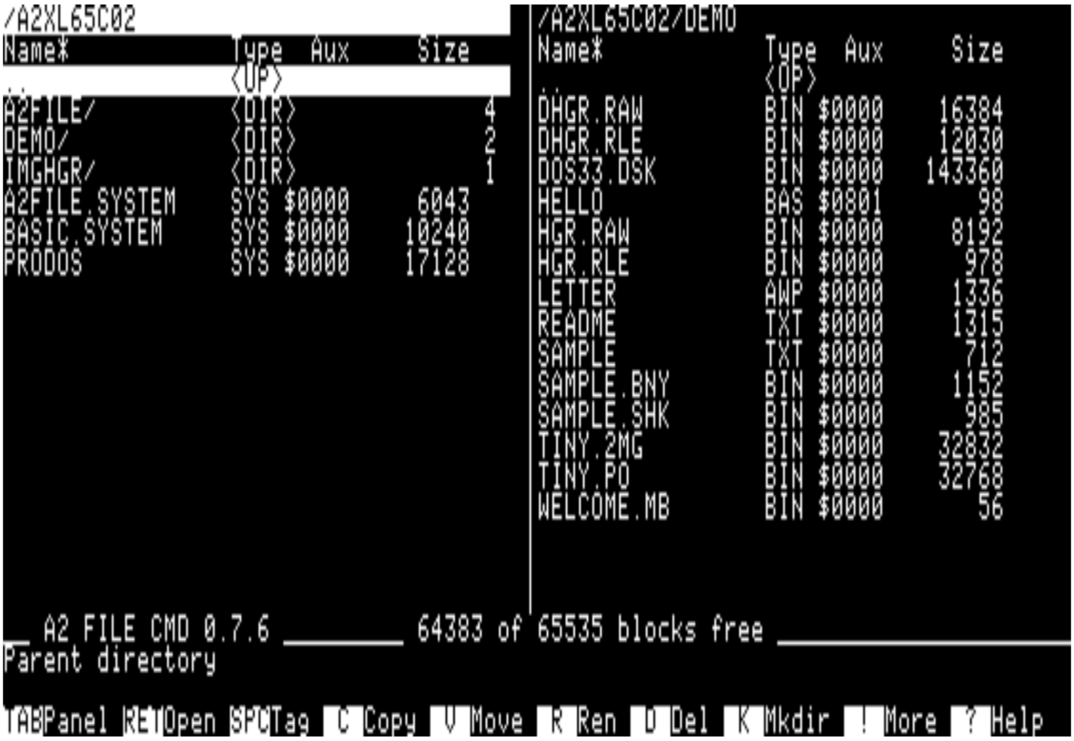
A2 FILE CMD

User guide · 0.8.0

6502 floppies by category · XL 6502 / 65C02



Contents

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English guide · Generated 12 September 2026

Version 0.8.0 — A two-panel ProDOS file manager for an Apple II with 128 KB and 80-column support. Free software by Arnaud Verhille, under GPL v3.

Start here

All floppies use **6502** code and run on an original or enhanced IIe. For XL, choose **6502** for an original IIe, or **65C02** for an enhanced IIe or //c with optional AppleMouse II support. Real-hardware validation on Apple //c, enhanced IIe and unenhanced IIe was confirmed successful by the maintainer on September 12, 2026. The IIgs has not been tested. The launcher checks the CPU and memory before starting and identifies the edition on its title screen.

Choose your disks

Edition	What to use
6502 floppies	BOOT plus whichever 140 KB category disks you need: FILES, MEDIA, DISKTOOLS, DEVTOOLS. These floppies also run on enhanced machines.
XL 6502 or 65C02	One bootable 32 MB `.2mg` with all 60 overlays, BASIC.SYSTEM, INTBASIC.SYSTEM and demonstration files. No companion disk is needed.

The download names include the CPU, category and version:

Role	Image name
Boot	`A2FILECMD-6502-BOOT-0.8.0.dsk`
Files	`A2FILECMD-6502-FILES-0.8.0.dsk`
Media	`A2FILECMD-6502-MEDIA-0.8.0.dsk`
Disk tools	`A2FILECMD-6502-DISKTOOLS-0.8.0.dsk`
Development tools	`A2FILECMD-6502-DEVTOOLS-0.8.0.dsk`
Complete, 6502	`A2FILECMD-6502-XL-0.8.0.2mg`
Complete, 65C02	`A2FILECMD-65C02-XL-0.8.0.2mg`

All floppies are 6502 and supplied as `.dsk` in DOS sector order; XL uses `.2mg`. Use the downloaded files directly: changing an extension does not convert an image. Each floppy image is 143,360 bytes. Check downloads against `SHA256SUMS-0.8.0.txt`; if all release files are together, run:

```
sha256sum -c SHA256SUMS-0.8.0.txt
```

Boot the image, or launch `A2FILE.SYSTEM` from a ProDOS selector. To install elsewhere, keep `A2FILE.SYSTEM` beside its complete `A2FILE/` directory. Do not mix `A2FILE.CODE` and native plugins from different builds.

Read the panels

The highlighted panel is active; **TAB** switches sides. Each panel shows names, file types, auxiliary types and sizes. The selected row is highlighted, a star marks a tagged file, and **L** marks a locked file. The header's star identifies the sort order. Below the panels are free space, selection details, messages and the key bar. **?** opens the help screen.

The program remembers panel directories, sorting and the active side in `A2FILE/A2FILE.CFG` when you quit. Saving reserves an exclusive temporary file, verifies every byte after closing it, retains the old file as a backup during replacement, and verifies the installed file before removing that backup. A warning lets you cancel quitting if saving fails. Existing `A2FILE.TMP` or `A2FILE.BAK` files are preserved for recovery. This is not power-fail atomicity. On the first XL start, the right panel opens `DEMO/`; try its text, pictures, music and sample archives.

The companion floppy and disk swaps

The companions group tools by function. Every disk also carries MENU and the full command catalog. The distribution is declared in `config/packages.mk`.

Category	Plugins	ProDOS volume
FILES	EDIT, SEARCH, AWP, BINARY2, UNSHRINK, CRC, FIND, FIXTYPES, GOTO, IDENT, MDVIEW, RENAME, SYNC, MOVE, TREE	`/A2FILES6502`
MEDIA	IMAGE, MUSIC, DGRVIEW, EXTASIE, PACKFOT, PAINT816, PURPLE, LZ4FH, PRINTSHOP, FONTVIEW, PT3	`/A2MEDIA6502`
DISKTOOLS	BOOTBLK, BLKVIEW, BLKEDIT, DISKCMP, IMGCONV, MKIMAGE, RESCUE, UNDELETE	`/A2DISKS6502`
DEVTOOLS	BASLIST, DISASM, INTBASIC listings, plus BASIC.SYSTEM and INTBASIC.SYSTEM runtimes	`/A2DEVTOOLS6502`

BOOT keeps the essential file manager and disk operations. Use the **same release** for all floppies. XL is complete; never replace its 65C02 native plugins with the 6502 companions.

With two Disk II drives, keep BOOT in **slot 6, drive 1** and put the required category in **slot 6, drive 2**. The **!** menu lists every command; an absent tool's description starts with the volume containing it.

With one drive, choose the tool normally. If a disk is missing, the prompt names the required volume, slot, drive and file. Press **1** or **2** to choose the drive, insert the named disk, then press **Return**. If the input file was on the removed disk, a second prompt asks for that disk. **Escape** cancels and returns to the panels. The chosen drive is remembered for the session; these plugin-loading prompts use slot 6.

BOOT is `/A2FC6502`; the complete disks are `/A2XL6502` and `/A2XL65C02`. The menu retains every command when companions are absent. Tools that need both source and destination online still require another drive or volume. **DISKCMP S** and **W** → **Copy** have their own single-drive exchange modes.

Protect files in /RAM

Copy anything important out of /RAM before displaying DHGR, extracting ShrinkIt, using disk-image operations or physically formatting a Disk II floppy. These operations use auxiliary memory and can rebuild `/RAM` empty. A warning explicitly says that ALL `/RAM` files will be lost and asks for consent **before** AUX is used. Declining preserves its contents. The viewer may ask even for a plain HGR file because the same viewer can browse subsequent DHGR pictures; plain HGR itself does not need the reconstruction. The message line reports a reconstruction afterwards. The foreground MB1 and PT3 players preserve `/RAM`.

Returning to a parent with **Escape** or **Return on ..** selects the child you just left, including when it lies beyond the first directory window. The child is located by its current name, so an obsolete index cannot select an unrelated entry. If it has disappeared, the parent opens at its beginning.

Keys

Key	Action
Up / Down	Previous / next entry.
Left / Right, **< / >**, **- / +**	Previous / next page; **[/]** jumps to first / last entry.
TAB / =	Switch panel / show the same directory in the other panel.
RETURN / ESC / /	Open / go up / list volumes. Return asks before running a program.
', then letter or digit	Jump to the next name with that initial.
SPACE / *	Toggle the selection's tag / invert all tags.
Ctrl-T / Ctrl-N / Ctrl-R	Tag all / untag all / reread both panels.
S / M	Change sorting / tag files absent or different in size/date in the other panel.
C / V	Copy / move tagged entries, otherwise the selection, to the other panel. Includes directories. Move deletes originals after copying.
D	Delete tagged entries or the selection, including directory contents, after confirmation.
R / K	Rename / create a directory. Names: letter first, then letters, digits or periods; maximum 15 characters.
A / L	Change hexadecimal type/auxtype / lock or unlock. Locked files refuse deletion and renaming.
T / H / I	Read text / hexadecimal / picture. T also lists BAS and AWP files.
E	Edit text; on a directory or `..`, create a text file.
X	Run a program after confirmation, replacing A2FC.
W / F / !	Disk-image operations / format / plugin menu.
? / 1 … 0 / Q	Help / key-bar buttons / quit to ProDOS after confirmation.

Copying preserves name, type and auxiliary type. For an existing target, choose **O** overwrite, **S** skip, **A** overwrite all or **N** overwrite none. Existing destination directories are filled in. Copying into the same or a nested source directory is refused. Files are read back and compared before a move can delete its source. Overwrites protect the old destination as `A2FC.BAK`; failures attempt to restore it. A leftover backup must be examined before another replacement. Archive and image extraction refuse existing output files instead of silently overwriting them.

Very deep trees can exceed the recursive working space. A2FC checks the remaining stack before directory I/O and refuses an unsafe traversal with `Directory unreadable or too large/deep.` Copy/move counts the tree before transferring it; directory deletion also scans it before the first removal. Copy or delete smaller subdirectories separately if needed.

Long operations display progress, including each item during recursive deletes. **ESC** interrupts; completed work remains, an incomplete ordinary file copy is removed, and the final message reports what was done. Read the result before removing a disk.

The mouse

On 65C02, an AppleMouse II supplements the keyboard. Click a row to select it; click the selected row again to open it. Click a path to go up, a column header to change sorting, or a key-bar button to invoke it. Viewers, the editor and prompts use the keyboard.

More tools in the ! menu

Select the item first, press **!**, choose a category and press **Return**, then choose its tool and press **Return**. Categories are Files, Images, Music, Disks, Programming, System, Archives and Other (third-party tools). **Escape** returns to the category list, then to the panels. Tools are sorted by name within each category. **Up/Down** moves a line, **Left/Right** moves six lines (stopping at the list ends), and a letter jumps to a matching initial. Questions on the penultimate line appear in inverse video while awaiting a response, including typed names, confirmations, conversion choices and disk swaps. Ordinary information and results remain in normal video. The following tools supplement the main keys and readers.

On BOOT and XL

Tool	Operation
COMPARE	Compare the selected file byte by byte with the same name in the other panel.
TXTCONV	C = CR, L = LF, D = CRLF, H = clear high bit, S = set it, T = expand tabs, A = transliterate UTF-8 accents; incomplete sequences become `?` without consuming following text. Write in place or to the other panel. In-place conversion preserves the source on incomplete reads or changed size and refuses an existing TXTCONV.TMP.
DATE	S sets date/time from `DDMMYYYYHHMM` (1940–2039); impossible dates are rejected. F stamps modification dates on tagged files or the selection. Creation dates stay unchanged; a hardware clock may replace the entered time.
VERIFY	Read tagged files (skip directories), the selection, or every block of a volume. Report processed files and errors; ESC cancels. No writes.
TAGPAT	Name patterns: `=` any string, `?` one character. Add comma-separated filters: `T04` TXT, `>2000` or `<2000` bytes, `D` modified today. T tags, U untags, X replaces tags.
VOLNAME	Rename a ProDOS volume and update the affected panel/program paths.
WIPE	F checks live directory/file references against the bitmap before zeroing free blocks. Unreadable, inconsistent or unsupported allocation is refused. W zeroes the whole volume after `ERASE`; the running program's volume is refused.

On category disks and XL

VOLINFO is on DISKTOOLS; the menu requests that disk when necessary. Menu categories describe tasks and do not require changing disks just to browse.

Tool	Operation
VOLINFO	Audit allocation and fragmentation. M = bitmap (`.` free, `#` used), F = selected file blocks, E = export to the other panel. N/P pages; ESC returns. No repairs.
SEARCH	Find text in the active directory and tag matching files, ignoring case. ESC cancels a long scan and keeps tags already found.
FIXTYPES	Set type/auxtype from suffixes on tagged files or the selection; optionally remove suffixes. Image suffixes and `.SYSTEM` stay.
GOTO	P opens a typed `/VOLUME/DIRECTORY` path (63 characters max; Delete/Left edits, ESC cancels). Nine favourites: A adds, D then a digit removes, M then two digits reorders, 1–9 jumps. Saved in `A2FILE/GOTO.CFG`.
FIND	Search the volume by name pattern; start with `"` to search contents, ignoring case. TAB sets type (T, two hex digits) and modification dates (D, inclusive YYYYMMDD, 1940–2039); A clears filters. Undated files are excluded by date filters. N shows the next 20 results; Return jumps there. V on a text result shows occurrence offsets (hex) and excerpts; N/Space continues, ESC returns.
BLKVIEW	Read device or image blocks: H hex/ASCII, D directory, I index, N/P block, Space page, G four-digit hex block, F find four bytes (8 hex digits), A find next, X extract blocks, ESC back. Source stays unchanged.
EXTASIE	View Extasie/Chat Mauve ProDOS `$F2` images. The original count/repeat stream is decoded into the HGR page; ESC returns to the panels. `Return` and `I` select EXTASIE automatically on both processors.
DISASM	Read BIN/SYS as assembly: N/Space next, P previous (last 64 pages), C 6502/65C02, G seven-digit file offset, L four-digit CPU load address, R start, E export, ESC back. BIN uses its auxtype; SYS starts at $2000.
CRC	Calculate CRC-32 for the selection or tagged files. Results appear in pages of 20; a key continues, ESC at a page boundary stops the batch.
IDENT	Identify content, UTF-8 or Apple text; statistics cover the first 512 bytes.
MDVIEW	Wrapped Markdown/text; no forward limit. Up: last 64 pages. R: restart.
RENAME	Batch prefix, suffix, extension replacement/removal or numbering. For example E then BAK sets `.BAK`. Conflicts are skipped.
IMGCONV	Convert PO/HDV, DSK/DO and 2MG into the other panel, preserving disk blocks. Unsupported 2MG formats, block counts exceeding 16 bits, and data ranges inside the header or beyond the source size are refused before destination access. Read or seek failures abort conversion and remove incomplete output.
BOOTBLK	Copy ProDOS boot blocks from the boot volume to another volume after confirmation. Saves both originals in main memory, verifies writes and restores both blocks on error. An incomplete restoration is reported explicitly; the backup does not survive a power cut.
UNDELETE	Browse deleted ProDOS entries. N skips; R recovers a validated candidate to another online volume. Existing names are refused.
DISKCMP	V compares online ProDOS volumes; I compares images; S compares two Disk II disks on one drive. Reports differing blocks and the first mismatch.
MKIMAGE	Create an empty ProDOS PO or 2MG: 140 KB, 800 KB, 2/4/8 MB or 32,767 blocks. New images are data volumes, without a boot program.

Tool	Operation
RESCUE	F recovers a file; V recovers a ProDOS volume. Uses up to 30 attempts per block, zero-fills unreadable chunks and writes a LOG. Destination must be another online volume.
SYNC	Recursively copy missing or newer files to the other panel after confirming direction. Destination-only files remain; copies are read back before replacement.
MOVE	Move marked files, or the selected entry without marks. Within a volume, move without copying its data blocks; locked sources are refused. Every path component must still be a directory. A full subdirectory grows if space is available; damaged parent references are refused before writing. Across volumes, copy and verify a file before deleting the source; existing destination names are refused, and a size mismatch preserves the source and removes the incomplete copy. Directories across volumes require V. Available on FILES and XL.
TREE	Show file sizes and cumulative directory totals. Space advances a page; ESC exits.

Recovery and comparison limits

UNDELETE never changes the source directory, indexes or bitmap. It checks retained pointers, block counts, free blocks and index halves swapped by ProDOS DESTROY. An interrupted DESTROY can leave a deleted entry with damaged allocation information: the deleted marker alone is insufficient. Reused, inconsistent or ambiguous candidates are refused. Standard seedling, sapling and tree files, including sparse files, are supported; deleted directories and resource forks are not. Inspect recovered data before relying on it.

RESCUE writes `BASE.REC` for a file. Whole-disk recovery writes raw ProDOS parts `BASE.P01`, `BASE.P02`, etc., up to 16,000 blocks each. Concatenate them in numeric order for a PO image; a single part can simply receive `.PO`. `BASE.LOG` records zero-filled chunks and completion or interruption. Partial output stays after cancellation or a write failure.

SYNC skips equal-date or newer destination files. A source with an unknown date does not replace an existing file. It uses `A2FC.SYNC` temporarily and `A2FC.BAK` for rollback, preserving pre-existing files with those names. Verification requires matching contents and exact length; a size mismatch preserves the source and existing destination and removes the temporary copy. If replacement fails, the original is restored where possible; keep any remaining `A2FC.BAK`. Overlapping directory trees are refused. SYNC and TREE support paths shorter than 64 bytes and up to 16 directory levels; read errors and unsupported resource forks produce an error/incomplete result.

DISKCMP S buffers two blocks per exchange, so a full comparison needs many swaps. Each prompt names the expected disk and slot/drive; **1/2** changes the drive, Return retries and Escape cancels. A read error or cancellation never produces an “identical” verdict. Image comparison supports PO/HDV, DSK/DO and ProDOS-order 2MG.

VOLINFO supports ProDOS files, both forks and directories up to 16 levels. Large volumes take longer; incomplete counts are unconfirmed. File lists show data, index, master and extended blocks, omitting sparse holes. Exports include the selected file and describe the scan before report creation; existing names are refused. Only complete exports end with `END REPORT`; partial files remain. **MKIMAGE** refuses existing names and removes cancelled/incomplete new images; 32,767 blocks is the maximum that fits in a single ProDOS image file.

DISASM shows file offsets, 16-bit CPU addresses, bytes and instructions. C changes decoding at the current offset and resets page history; 65C02 includes Rockwell/WDC extensions. Unknown/truncated instructions appear as `.BYTE`. E exports from the current offset to EOF as a new TXT file in the other panel, using the displayed CPU and load address. Existing names are refused. ESC cancels; errors or cancellation keep partial output. Exporting preserves the current page.

BLKVIEW F searches forward from the current block, including matches across block boundaries; A continues without wrapping. X extracts from the current block: enter a four-digit hex count (maximum `7FFF`) and a new filename in the other panel. Device sources require another destination volume. Errors retain partial output.

Disk writes and copies automatically read back every written block. A readback error stops at the first failing block. On one drive, each prompt identifies SOURCE or TARGET copy, the source volume and the slot/drive.

Reading files and pictures

Text, hexadecimal and AppleWorks

Use **T** for text, **H** for hexadecimal. **Space**, **Return** or **Down** advances a page; **B** or **Up** goes back; **R** restarts TEXT at page one; **R** also restarts BASLIST and AppleWorks; **ESC** returns to the panels. The text reader clips lines beyond 80 columns and remembers up to 96 page starts. Use **MDVIEW** for wrapped text. The hex reader shows addresses, bytes and text. HEX: **G** goes to a seven-digit file offset; **R/E** first/last.

T on a BAS file lists Applesoft line numbers and keywords. T or Return on an AWP document reads AppleWorks word-processor text; formatting commands are omitted and tabs expanded. AppleWorks databases and spreadsheets are not supported.

Pictures

Return and **I** use the same picture-format selection on 6502 and 65C02. The viewer is loaded automatically. Explicit packed ProDOS types take precedence; otherwise the first eight bytes identify DGR and HGRR/DHRR without requiring a filename suffix or a ProDOS image type. A failed identification read or close stops opening the file.

Format	Identification	Viewer
Extasie / Chat Mauve	ProDOS type `$F2`	EXTASIE
Packed FOT / PackBytes	Type `$08`, auxiliary `$4000` or `$4001`	PACKFOT
LZ4FH compressed HGR	Type `$08`, auxiliary `$8066`	LZ4FH
Print Shop monochrome clip art	BIN, auxiliary `$4800/$5800/$6800/$7800`, 572 or 576 bytes	PRINTSHOP
MGTK / Apple II Desktop font	Type `$07`	FONTVIEW
Purplesoft GRLOAD pair	Matching `.FOT01` and `.FOT02`, 8 KB each	PURPLE
Packed 816/Paint	Type `$06`, auxiliary `$E001` or `$E002`	PAINT816
Lo-res page	Type `$06` or `$08`, auxiliary `$0400`, 1–2,048 bytes	DGRVIEW
DGR with header	`DGR` signature; the viewer validates the header and payload	DGRVIEW
HGRR / DHRR RLE	Version 1 header, or type `$06/$08` with `.RLE` suffix	IMAGE

Format	Identification	Viewer
Raw HGR / DHGR	Type `$06` or `$08`, raw page size	IMAGE

Raw HGR accepts 8,192 or 8,184 bytes; raw DHGR accepts 16,384 or 16,376, auxiliary plane first. **Left/Right** browses the raw/RLE album and skips formats handled by other viewers. **Escape** returns from every picture viewer. For an unmarked lo-res screen or sprite (BIN/FOT, 1–2,048 bytes), press **I**: DGRVIEW opens and asks for the width when needed. **Return** keeps ordinary small binaries in hex because a sprite has no signature. **H** explicitly opens hex, including for a picture. Unsupported binary formats retain the hex fallback.

FONTVIEW displays the glyphs in code order, sixteen per row. It accepts MGTK fonts with one or two seven-bit columns, up to 128 glyphs and 22 rows per glyph. PRINTSHOP displays 88 × 52 clip art at its 2 × 3 display scale. LZ4FH, PRINTSHOP and FONTVIEW use only main memory and preserve `/RAM`. They reject truncated data and report read or close errors.

Left/Right browse the previous/next file handled by the same specialized viewer: Extasie, PACKFOT, 816/Paint, Purplesoft, DGRVIEW, FONTVIEW, LZ4FH and PRINTSHOP. The directory's displayed order is used, including across large-directory windows. At either end the arrow does nothing; Escape returns to the panels. A viewer that uses AUX asks once before the first AUX write in a browsing session. Left/Right keeps that consent; leaving the viewer clears it.

PURPLE opens either member of a Purplesoft GRLOAD/GRSAVE pair. Keep both files in the same directory, with the same basename: `.FOT01` holds the auxiliary plane and the saved mode, `.FOT02` holds the main plane. Arrows skip the companion file. The first opening asks for AUX consent; malformed or incomplete pairs are refused. Extended EVE modes (COL280A/B, CP280 and special HGR modes) require compatible EVE hardware/emulation for correct colours; ordinary RGB cards support the usual COL140/BW560 modes. This reader does not yet accept Pascal GLOAD single-file pictures.

T on an Integer BASIC file (`$FA`), including `WOZ.BREAKOUT` and `APPLEVISION`, opens INTBASIC.PLG to list its source. **Return/X** executes it through INTBASIC.SYSTEM after confirmation.

Destructive AUX use requires prior consent. Opening a viewer again starts a new session and asks again; declining preserves `/RAM`. With floppies, specialized picture viewers are on **MEDIA**, and BOOT holds the internal `OPEN.PLG` dispatcher. Keep it with the matching program build.

The Mockingboard music

Return on an MB1 `.MB` stream opens the foreground MUSIC overlay on MEDIA (maximum 4,096 bytes, six voices). **P** pauses/resumes; **Escape** returns to the panels. Playback also returns at the stream's end. The complete file is read, closed and validated before playback. The player uses main RAM and preserves `/RAM`; an absent Mockingboard is reported.

On Apple //c, a Mockingboard 4c uses the internal connector and answers at `$C400–$C4FF`. The same automatic probe supports it; both MB1 and PT3 use the detected address page. In POM2, enable the card in the //c configuration and use a build containing its Mockingboard 4c support. A plain //c without the card reports its absence.

In both music players, **Left/Right** select the previous/next tune of the same type in the same directory (MB1 stays with MB1, PT3 with PT3), including across large-directory windows. An arrow without a neighbour does nothing. Changing tracks stops the old output and starts the new track unpaused.

Return on a `.PT3` module opens the foreground ProTracker 3 player on MEDIA. Ordinary modules use three voices on the first AY chip. Standard TurboSound `02TS` containers use both AY chips for two independent modules and six voices; playback ends once both modules have finished. **P** pauses/resumes; **Escape** stops and returns to the panels. Playback also stops at the end of the song. A missing card is reported without starting playback. The screen shows the module's title and artist/credit field, then **Player: Vince Weaver - A2FC adapter**. Empty titles use the filename; empty credits show **Not specified**. Fixed-width header fields are bounded and control characters are removed from the display; the module is not modified.

PT3 accepts modules up to **65,535 bytes** using a main-RAM page cache and preserves `/RAM`. The source stays open read-only throughout playback: keep its disk mounted. Slow storage can delay playback when an uncached page is needed. **Esc** also cancels the initial file scan. The existing filtered corpus in `media/pt3/MUSIC/<ARTIST>/` and on `/GISTDATA/MUSIC/<ARTIST>/` contains 5,507 files in 449 artist folders; it has not been regenerated for the larger limit. Eight starter modules remain in `media/pt3/` and on the `A2FC-PT3.po` volume. Frequency tables 0–3 are supported, including their older PT3 3.0–3.3 variants. Multiple deferred special effects within one channel/row are refused. TurboSound accepts the standard two-PT3 container with a 16-byte `02TS` footer; the 65,535-byte limit includes both modules and the footer. Other multi-chip container variants are unsupported. At 1 MHz, sparse dual modules can exceed the small cache and slow down even on a hard disk; this implementation does not guarantee 50 Hz for every file. Invalid data, stream pointers, or read/seek/close errors stop loading or playback and return to the panels. Source URLs are listed in `media/pt3/MUSIC/SOURCES.TXT` and at the end of SAMPLE-MEDIA.md.

Moving marked files with MOVE

When files are marked, **!** → **Files** → **MOVE** moves the marked files after one confirmation. It reserves and verifies `A2MOVE.LST` in the destination before moving any source. An existing list is preserved and blocks the operation. Each cross-volume copy is verified completely before its source is removed. The batch stops on the first error or Escape; completed moves remain complete, and pending visible source files are marked again by name. Destination marks are cleared. Marked directories are refused; the existing single-entry MOVE remains available without marks. A retained list is reported for manual review. No auxiliary memory is used.

The text editor

E edits a file; on a directory or `..`, it creates one. The editor holds up to 5,104 bytes, uses CR line endings and strips the high bit on loading. Long lines do not wrap. A star on the status bar means unsaved changes.

Arrows move; **Delete / Ctrl-D** erase left / right. **Ctrl-A / Ctrl-E** go to line start / end; **Ctrl-P / Ctrl-N** changes page; **Ctrl-T / Ctrl-B** goes to text start / end. **Return** splits a line; **Tab** inserts four spaces.

ESC opens the menu: **S** save, **X** save and exit, **Q** quit without saving (confirm if changed), **ESC** continue editing.

Saving preserves the file's ProDOS type and auxiliary type. It writes and reads back `A2FC.EDIT` before installing it, using `A2FC.ED.BAK` to protect the previous file during renaming. Extra free space is required. If either name already exists, examine/recover it before removing it; A2FC never overwrites a previous recovery file.

Disk images and formatting

Disk images

W opens three operations: write the selected image to a formatted target disk, read a disk into a new image in the active directory, or copy one disk to another. Supported containers are PO, DSK/DO and 2MG.

For a one-drive copy, choose the same drive for source and destination, then follow the disk prompts. Writing requires **ERASE** in capitals and refuses the running program's disk. Check the named target before confirming. Escape cancels before writing. Disk-image operations can clear `/RAM`.

A disk image as a folder

Return on PO, DSK/DO or 2MG opens a supported ProDOS or DOS 3.3 image read-only. Navigate with Return and Escape; Escape at its root leaves the image. **C** extracts tagged files, or the selection, to a real ProDOS directory in the other panel, preserving type and auxiliary type.

ProDOS image extraction supports files up to 128 KB (seedling/sapling). Enter subdirectories and extract their files individually; recursive extraction and writing into an image are not supported.

A real DOS 3.3 disk appears in **/** as `DOS 3.3`, with its slot and drive. Return opens its catalog; C extracts files to ProDOS. Applesoft, Integer and binary DOS headers are removed during extraction. DOS 3.3 access is read-only.

Formatting a disk

Press **F** or choose **FORMAT** in **!**. Select the drive, enter a volume name, then type **ERASE** and press Return. The list shows slot, drive, volume and capacity. The program's own volume is marked **IN USE** and cannot be formatted. Escape or a different confirmation word cancels.

Disk II formatting clears `/RAM`; move its files elsewhere first. It is refused if A2FC itself runs from `/RAM`. Formatting another block device does not require that extra RAM reset. The result reports success or an error; Escape returns directly to the panels.

Archives and programs

Unpacking archives

Set the destination in the other panel, select the archive, then use **!**:

Tool	Supported archives
UNSHRINK	ShrinkIt `.SHK`: stored data, LZW/1 and LZW/2. Files retain ProDOS type and auxiliary type; names are adapted to ProDOS. Disk-image members become PO files. Resource forks and comments are skipped.
BINARY2	Binary II `.BNY`/`.BQY`: extracts members with their names and attributes. Directory entries are skipped. Compressed members may need a second extraction with UNSHRINK.

ShrinkIt extraction clears `/RAM` and refuses it as a destination. Keep archives and recovered files on another volume.

Running BASIC and machine-language programs

Return runs BAS, INT or SYS after confirmation; **X** also runs BIN. The launched program replaces A2FC; it does not automatically return. A BIN loads at its auxiliary address, which must be between `$0800` and `$BAFF`.

Applesoft (`$FC`) requires `BASIC.SYSTEM`; Integer BASIC (`$FA`) requires `INTBASIC.SYSTEM` v0.9. Both are supplied on DEVTOOLS and XL; the runtime is chosen automatically. **T** lists either BASIC source without executing it. A2FC searches the program's volume, its own volume and the companion, and names DEVTOOLS when the runtime disk is absent. An unreadable or incompatible runtime is refused. With one floppy drive, keep the BASIC program on another online volume so it remains readable after the runtime loads.

Launch checks use the file's actual size and keep the load below `$BB00`, where the loader's ProDOS I/O buffer starts. If BOOT is full and preferences cannot be saved, A2FC asks whether to run anyway; it does not silently discard that failure. Integer BASIC returns to the ProDOS selector at program end. Its upstream runtime supports a subset of DOS commands; see INTBASIC.SYSTEM usage and provenance.

From the Applesoft `]` prompt, reinsert BOOT if needed and enter:

```
-/A2FC6502/A2FILE.SYSTEM
```

Use `/A2FC65C02/` for the enhanced BOOT, `/A2XL6502/` or `/A2XL65C02/` for XL, or the actual installation path. `-A2FILE.SYSTEM` also works when the current prefix is its directory. BAS paths over 46 characters use a fallback launch method; use a shorter path if launch fails.

VDrive: two volumes over the serial line

A detected Super Serial Card or //c serial port can expose two remote ProDOS volumes at **115,200 bps**. Slot 2 is tried first. The status line identifies the serial card and assigned drive slots; the remote volumes then work like local ones for browsing and copying.

Use ADTPro's virtual-drive server, `veserver.py` or `surl-server`; the host supplies date/time during reads. Disconnection produces an I/O error. The driver is removed on quit or program launch. Tested in POM2; real Super Serial Card and //c operation remains unverified.

Limits and troubleshooting

Symptom or limit	What to do
CPU or memory refusal at startup	Use the matching CPU build, with 128 KB and 80-column support.
Missing or stale plugin	Insert matching BOOT/category disks from the same release. Keep A2FILE.CODE and its native plugins together.
GOTO.TMP or GOTO.BAK remains	If GOTO.CFG is missing, rename GOTO.BAK to GOTO.CFG. Inspect remaining TMP/BAK files before deleting them.
Disk changed but old contents remain	Press Ctrl-R to reread both panels.
Run failed: file not found	Check the selected program, its path and BASIC.SYSTEM for BAS files.
Image cannot be opened	Check its actual format; renaming PO to DSK does not convert it. Some file/image features are unsupported.
Directory exceeds 139 entries	A2FC uses windows in disk order, without sorting. Cross a window edge to continue.
Very deep directory copy refused	Simplify the tree or copy smaller subdirectories; recursive operations have bounded working space.
Incomplete scan or recovery	Read the reported limits or errors; inspect recovered data and keep originals/backups.

`make disk` builds both CPU families; `make test` runs host checks. Development: README, `sdk/README.md`, `bench/README.md`. Remaining work: `TODO.md`.

Credits, inspirations and reused code

A2FileCmd is original GPL v3 software by **Arnaud Verhille**. The links below identify the projects that shaped its interface, supplied technical references, or contributed code patterns. They are listed so that the provenance of every borrowed or adapted fragment is easy to check.

cc65 — Oliver Schmidt Toolchain and Apple II startup code. `src/crt0.s` and `src/crt0_loader.s` retain the V2.19 provenance notice; local loader changes are marked there. URL: cc65 repository · Apple II startup source

Colin Leroy — a2tools / Ammonoid The serial virtual-drive glue in `src/vsdrive.s` follows the documented a2tools approach; the source comment names the corresponding file and author. URL: a2tools repository · Ammonoid releases

Jerry Hewett and Gary Desrochers — Hyper-FORMAT The Disk II ProDOS formatter descends from their public-domain routines, carried by ADTPro and credited in `src/format.c` and `src/format_diskii.s`. URL: ADTPro source tree

ProDOS 8 documentation File-system structures, MLI calls, allocation blocks and path rules follow the published Apple II technical references. URL: ProDOS 8 technical information

A2Command The Apple II two-panel file-manager model and command-oriented disk workflow are explicit design inspirations. URL: A2Command archive

Norton Commander The two-panel layout, selection marks and bottom key bar are interface inspirations. URL: Norton Commander archive

ShrinkIt / NuFX and Binary II `UNSHRINK` implements documented LZW/1 and LZW/2 formats; `BINARY2` follows the public member-header format. No archive executable code is bundled. URLs: NuFX notes · NuLib format library

POM2 — Arnaud Verhille Separate emulator project used for repeatable Apple IIe, //c and disk-device verification. URL: POM2 repository

ADTPro Virtual-drive and disk-transfer workflows support moving the supplied `.dsk` images to real hardware. URL: ADTPro project

Reading the source notices

The repository keeps attribution next to the affected implementation:

- `src/crt0.s` and `src/crt0_loader.s` identify the cc65 startup source and describe the local staging changes.
- `src/vsdrive.s` identifies the a2tools file and explains the adapted serial entry points.
- `src/format.c` and `src/format_diskii.s` identify the Hyper-FORMAT lineage and separate the original formatter work from A2FileCmd integration.
- `src/unshrink.s` describes the documented archive algorithms; its decoder is an independent implementation written for the overlay memory limits.

The project does not copy Norton Commander, A2Command, ProDOS or ADTPro executables. Their interfaces, manuals and protocols are references or inspirations. The generated demo files are also original: `tools/mkdemo.py` creates the pictures, music, text and sample archives during the build. When redistributing a modified build, keep this section, the source notices and `LICENSE` with the program so the attribution remains visible.

Reference index

For readers who want to compare the implementation with its references:

- A2FileCmd source and issue tracker
- cc65 documentation
- cc65 Apple II library sources
- a2tools source and releases
- ADTPro documentation
- ADTPro source
- ProDOS 8 technical reference
- CiderPress II format notes

- NuLib library and Binary II references
- Apple II FAQ archive
- A2Command archive mirror
- Norton Commander archive
- POM2 emulator
- ZX-Art AY / PT3 catalogue
- ZX-Art PT3 API
- ABSTRACT (Ra, 1999) PT3
- dh2020rt (EA, 2020) PT3
- Music (VAD, 2002) PT3
- Old Skool For Demodulation (EA, 2020) PT3
- realtime blast (EA, 2026) PT3
- Yazzie: final theme (nq, 2019) PT3
- ana_ng.pt3 in dos33fsprogs
- mA2E_3.pt3 in dos33fsprogs
- Vortex Tracker II
- zxtunes.com author list
- zxtunes.com Macros archive
- zxtunes.com Korund archive
- Vince Weaver pt3_lib

These URLs were checked when this edition was prepared. A historical archive may move or disappear; the repository copies the relevant attribution and source-path information so the record remains useful if a mirror changes.

Only the fragments identified above are derived from external code or documented routines. The A2FileCmd overlays, ProDOS walkers, image readers, tests, demo data and user interface were written for this project. Generated demo files are created by `tools/mkdemo.py`; they are not copied from the inspiration projects. See the repository source comments and `LICENSE` for the complete copyright and redistribution terms.

See Data safety for recovery-file names, failure coverage and the limits of recovery after interrupted physical writes.